

Based on 49 sources

Method Name	Purpose	Steps or Instructions	Expected Outcome	Additional Notes
HALIRA (Hegelian Algebraic Loop for Irreducible Recursive Architecture)	To shift from symbolic generation to an engineered architecture for unified field theory by modeling intelligence as a recursive process acting upon filtration layers.	1. Organize system states by 'contradiction order' (K_p) using Contrarecursive Filtration. 2. Employ a Metamorphic Boundary Operator (∂_H) satisfying $\partial_H^2 = id$. 3. Use a Spectral Sequence Engine where differential d_r maps paradoxes down through filtration levels.	Achievement of Terminal Stability where a paradox cannot be further reduced; establishment of a computable link between algebraic topology, logic, and number theory.	Replaces standard metrics with a metric of instability; utilizes Homology to track 'barcodes'.
Cayley-Stiefel projection	To instantiate an orthogonality-preserving mixer for structured skip connections, preventing eigenvalue contraction and eigenspace misalignment to maintain	1. Skew-symmetrize unconstrained matrix: $W = \tilde{H} - \tilde{H}^T$. 2. Initialize: $Y_0 = I_n + \alpha W$. 3. Iterate s times: $Y_{i+1} = I_n + (\alpha/2)W(I_n + Y_i)$.	Norm-preserving mixing with exact gradients and negligible overhead; achieves dynamical isometry.	In practice, $s = 10$ suffice; achieve deviation $< 10^{-6}$.

